Dataset integrity & feature audit

Actual UCI Extension of Z-Alizadeh Sani Dataset; no synthetic patient records added.

Source records	**303**
Source columns	**59**
Candidate predictors after target exclusion	**55**
Missing data / duplicate rows	**0 / 0**
Constant feature: Exertional CP	**1**
Conflicting vessel-vs-Cath label in source	**1 row**

Target leakage guard

Cath (overall CAD), LAD, LCX and RCA are excluded from all prediction input matrices. The label disagreement in Excel row 95 was reported, not silently corrected.

Redundancy evidence

- BMI = Weight / (Length / 100)^2 for all 303 records (maximum absolute error: 0).
- Spearman Lymph vs Neut: -0.912; Weight vs BMI: +0.720.
- DM vs fasting blood sugar: +0.644; HTN vs measured BP: +0.616.

Feature strategy & validation

Selection procedures fit only on training folds to avoid optimistic validation.

The 10-question rapid CAD workflow

Age, Sex, BMI, Diabetes, Hypertension, Current smoking, Chest-pain presentation, Shortness of breath, Blood pressure and Pulse rate. Chest pain is captured by one dropdown mapped to three mutually exclusive source columns. Function Class is excluded.

V4 reduced model: observed nested cross-validation

Pooled out-of-fold CAD ROC-AUC

0.899

5 outer folds; 3 inner folds

Logistic Regression vs Random Forest

Algorithm selection inside each outer train fold

Earlier exploratory tier experiment (distinct CV procedure)

Scores above come from nested validation on the same historical dataset. They are not independent external evidence, and differing experiments should not be interpreted as a direct head-to-head test.

Engineering decisions & responsible interpretation

Ablations and patient-level explanations are reproducible and model-specific.

Handcrafted feature ablation — five-fold mean ROC-AUC

Logistic Regression	No interactions	**0.8990**
Logistic Regression	With interactions	**0.9003**
Random Forest	No interactions	**0.9052**
Random Forest	With interactions	**0.9034**

Interactions tested: Age × HTN; DM × HTN; Age × typical chest pain; BMI × DM. These did not provide consistent improvement and were not retained solely for complexity.

Explainability implementation

For the 10-question logistic model, SHAP LinearExplainer reports local contributions in log-odds. Full Gradient Boosting and Random Forest classifiers use TreeExplainer where available. RBF-SVM uses a labeled one-feature reference-substitution sensitivity analysis (not SHAP). These explanations quantify model behavior, not disease causes.

Age-band comparison

For age 45, the reference band is 40–49: 53 patient records, of which 28 have CAD labels. Cohort means and label proportions are calculated from this referral-selected dataset. Bins with fewer than 10 records do not display detailed comparison values. Individual model scores are not equated with cohort prevalence.

Limitations and reproducibility

The dataset contains 303 clinically selected records, one inconsistent target row, and limited generalizability. Probability calibration and external clinical validation have not been established. Training code, CSV feature-audit evidence and nested-CV logs are included alongside the project. The 3D heart is reference anatomy, not patient-specific angiography.